Customer Churn Prediction for Subscription Businesses

Project Report | Data Science | Persevex Project Dashboard

Abstract

This report presents an end-to-end machine learning system that predicts which subscription customers are likely to churn in the next 30 days and explains why. It combines demographic, billing, product-usage and support data into a unified feature matrix, trains class-imbalance-aware models (XGBoost, Random Forest), evaluates them with a financial cost framework, and explains every prediction with SHAP so retention teams can act early.

Project Links

Resource	Link
GitHub: Churn Prediction source code	https://github.com/ashik-code-design/CUSTOMER_CHURN_PREDICTION_SUBSCRIPTION
GitHub: Automated Survey Analysis	https://github.com/ashik-code-design/automated-survey-analysis
Live Streamlit app (demo)	https://customer-churn-prediction-subscription.streamlit.app/

1. Executive Summary

Recurring revenue drives valuation in subscription businesses (SaaS, telecom, streaming, e-commerce), but churn directly erodes it. Because customer acquisition costs are typically 5 to 25 times higher than retention costs, reducing churn is the most effective lever for increasing Customer Lifetime Value (CLV) and profitability. The system forecasts churn before it happens and surfaces the drivers, enabling Customer Success and Marketing teams to run targeted, automated retention campaigns.

2. Problem Statement and Objectives

Churn destroys future compounding growth: a business growing 5% monthly with 1% churn ends up far larger after five years than one with 3% churn.

- **Voluntary churn:** customer cancels due to dissatisfaction, low perceived value or competitors.
- **Involuntary churn:** subscription lapses from payment failures, expired cards or billing errors.
- **Reactive trap:** exit surveys arrive after the relationship has ended; the goal is proactive intervention.

Project objectives

- **360-degree data pipeline:** unify CRM, billing and product analytics in one feature store.
- **Predictive modeling:** train XGBoost and Random Forest on severely imbalanced data.
- **Explainability (XAI):** use SHAP for human-readable reasons behind each prediction.
- **Operational integration:** write scores to Salesforce, HubSpot and Intercom to trigger workflows.

3. Data Architecture and Feature Engineering

Data domain	Examples
Demographic and account	Industry, company size, geography, subscription tier, tenure
Financial and billing	MRR, billing frequency, discount usage, payment delays
Product telemetry	DAU, feature adoption, session length, time-to-value
Support and sentiment	Open tickets, resolution time, NPS, chat sentiment

Engineered features capture behavioral trends rather than static states:

- **RFM analysis:** recency, frequency and monetary value of engagement.
- **Velocity and delta features:** 7-day vs 30-day moving averages; for example a 75% drop in weekly logins is a strong red flag.
- **Support Friction Index:** combines ticket frequency with resolution time; correlates with voluntary churn.
- **Inactivity streaks:** consecutive days without login, weighted exponentially.

4. Exploratory Data Analysis

Cohort analysis groups customers by acquisition month and visualizes retention curves as heatmaps, exposing macro issues such as a promotional cohort churning 40% more by month 6.

Survival analysis (Kaplan-Meier) estimates S(t), the probability a customer survives beyond time t. It reveals critical drop-off points, such as after a 14-day free trial or before annual renewal, and so guides the timing of interventions.

5. Modeling Strategy

Churn is framed as supervised binary classification: using data at time T, predict whether the customer churns by T + 30 days.

Handling class imbalance

- **SMOTE:** synthetic minority-class examples generated from feature-space nearest neighbors.
- **Algorithmic penalties:** scale_pos_weight in tree models to penalize missed churners (false negatives).

Algorithms compared

Model	Role
Logistic Regression	Interpretable linear baseline
Random Forest	Handles non-linear, high-dimensional data with low overfitting risk
XGBoost / LightGBM	Champion models; sequential shallow trees correct earlier residuals and capture nested behavioral patterns

6. Evaluation and Expected Value Framework

Accuracy is misleading: with a 5% churn rate, a model that always predicts "will not churn" scores 95% yet is useless. Evaluation therefore focuses on:

- **Recall:** share of actual churners found; maximizes intervention opportunities.
- **Precision:** share of flagged customers who truly intended to leave; protects retention budget.
- **PR-AUC** (prioritized over ROC-AUC) for its sensitivity to minority-class performance.

The decision threshold is tuned with a financial cost matrix: a false positive costs, for example, a $50 discount to a customer who would have stayed, while a false negative costs a customer with a $5,000 CLV. The threshold maximizes net financial retention, not statistical symmetry.

7. Interpretability with SHAP

A bare prediction such as "87% churn probability" is not actionable. SHAP, rooted in cooperative game theory, computes each feature's marginal contribution to a prediction.

- **Global:** summary plots show macro drivers, e.g. a drop in feature adoption over 14 days as the top predictor.
- **Local:** per-customer waterfall, e.g. base risk 5%, +40% no login for 21 days, +25% critical ticket open over 48 hours, -10% annual plan = 60% final risk.

8. Deployment and MLOps

- **Batch scoring:** Apache Airflow runs a nightly ETL that builds rolling-window features; MLflow serves the latest validated XGBoost model.
- **Reverse ETL:** user_id, churn_probability, risk_tier and top-3 churn reasons go to Snowflake/PostgreSQL, then to Salesforce, Zendesk and Marketo via Hightouch or Census.
- **Drift monitoring:** feature drift (e.g. a UI bug changes login patterns) and concept drift (e.g. tickets stop predicting churn); PR-AUC below threshold triggers automated retraining.

9. From Predictions to Business Strategy

Segment	Action
High risk / High CLV (White Glove)	Priority CRM alert and a senior CSM call tailored with SHAP reasons, e.g. a training session for low adoption
Medium risk / Low CLV (Automated)	Email drip campaigns with content, webinars or a dynamic 15% discount
Involuntary churn risk	SMS or in-app prompts to update payment details, e.g. card expiring in 15 days

10. Future Scope: Uplift Modeling

Standard models suffer from the "Sleeping Dog" problem, where a retention message prompts cancellation. Uplift modeling uses A/B test data to separate four groups:

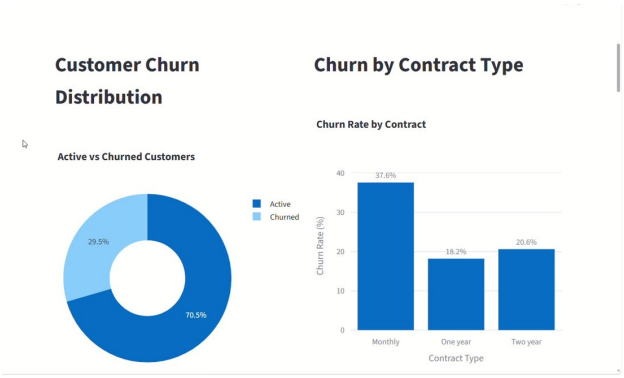
Group	Behavior	Action
Persuadables	Stay only if contacted	Target
Sure Things	Stay regardless	Skip discount
Lost Causes	Leave regardless	Skip effort
Sleeping Dogs	Leave because of contact	Do not touch

11. Conclusion

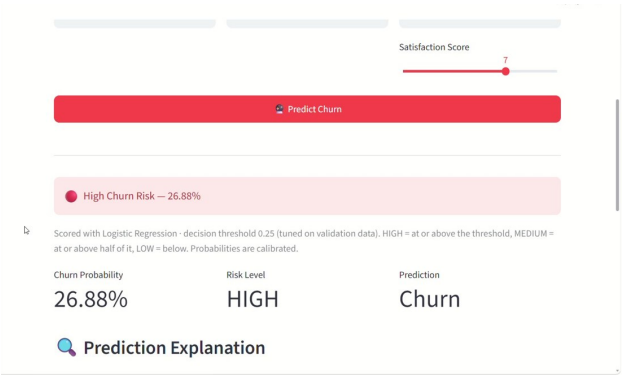
By fusing data engineering, advanced machine learning and explainability, the system turns raw telemetry into targeted retention action. It replaces intuition with predictive mathematics, helping organizations defend their customer base, optimize retention spend and protect long-term compounding growth.

12. Live Application Screenshots

Screenshots from the deployed Streamlit app show the dashboard, prediction, model analysis, SHAP explainability and survival views described in this report.



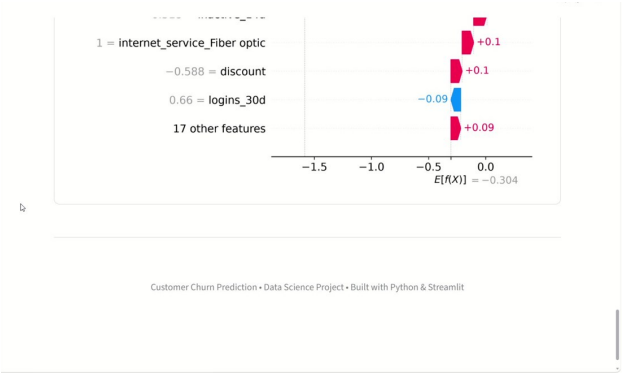
Business dashboard: churn distribution and churn rate by contract type



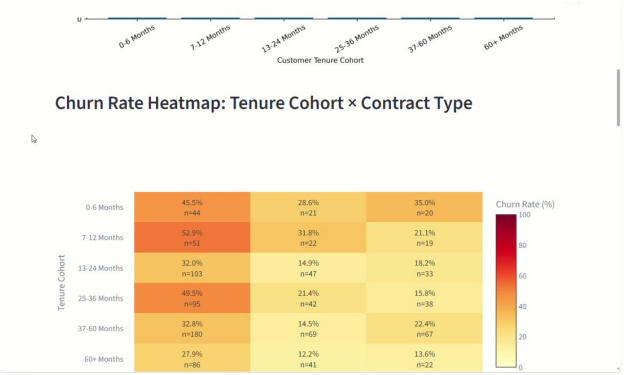
Churn prediction: 26.88% probability, HIGH risk



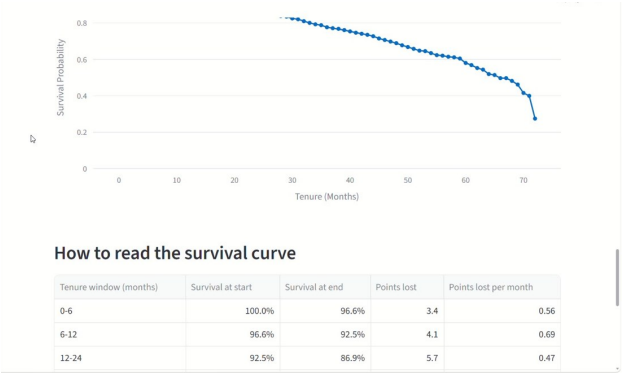
Model analysis: XGBoost has the highest recall (0.814)



SHAP explainability: feature contributions for one customer



Customer survival: churn heatmap by tenure cohort and contract



Survival curve: probability of staying over tenure

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